

Jeffrey Li

Senior AI Product Engineer |

Generative AI • Machine Learning • Data Engineering • Full-Stack Systems

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PROFILE

Innovative Senior AI Product Engineer with deep expertise in Generative AI, Large Language Models, multimodal systems (text–audio–video), and scalable ML platforms. Over a decade of experience building end-to-end AI products across healthcare, finance, customer experience, and enterprise AI. Proven track record of designing agentic workflows, RAG systems, multimodal embeddings, and production-grade LLM applications that serve millions of users and deliver measurable business impact.

Highly skilled in agent development, distributed model training, fine-tuning (LoRA/QLoRA/DPO/ORPO), semantic search, conversational AI, video/audio intelligence, MLOps, and large-scale data engineering. Adept at turning ambiguous business problems into robust AI solutions by combining deep technical expertise, product intuition, and customer-centric design.

Recognized for driving cross-functional collaboration, accelerating AI adoption, and transforming complex research innovation into high-performing, secure, and reliable enterprise-grade AI products. Passionate about building real-world AI systems, reducing hallucinations, improving model efficiency, and shaping the next generation of LLM-powered intelligent applications.

SKILLS

Generative AI & LLMs

LLM development, RAG 2.0, retrieval orchestration, LLM fine-tuning (LoRA, QLoRA, DoRA, DPO, ORPO), prompt engineering, LLM evaluation (Ragas, DeepEval), model compression (quantization, distillation), tool-use & function-calling, LLM safety, red-teaming, multimodal LLMs (text-audio-video), agentic systems, long-context optimization.

Machine Learning & Deep Learning

Transformers, sequence models, classical ML (XGBoost, LightGBM, CatBoost), graph ML, recommendation systems, ranking & search, anomaly detection, representation learning, contrastive learning, time-series ML, probabilistic modeling, active learning, weak supervision (Snorkel), optimization/regularization, self-supervised learning.

Multimodal AI (Text, Audio, Video, Vision)

Vision transformers (ViT, CLIP, SAM2), video transformers, diffusion models, 3D CNNs, Speech/ASR/TTS (Whisper, MMS, Kaldi), speaker diarization, audio embeddings (OpenSMILE, librosa), video feature extraction, semantic video indexing, multimodal embedding alignment.

Conversational AI / NLP

Intent classification, NER, summarization, conversational search, semantic retrieval, ranking, dialog state tracking, sentiment/emotion detection, QA systems, text analytics, customer-care automation, persona modeling, contact-center analytics.

Large-Scale ML Systems

Distributed training (FSDP, DeepSpeed, Megatron-LM, Ray), data-parallel & model-parallel strategies, GPU/TPU optimization, TensorRT-LLM, vLLM, Triton kernels, batch inference, streaming ML pipelines, high-throughput microservices (gRPC/FastAPI).

Data Engineering & AI Data Infrastructure

Spark, PySpark, Airflow, Dagster, Kafka, Flink, dbt, Delta Lake, Iceberg, Snowflake, BigQuery, real-time ETL, feature stores (Feast, Tecton), vector databases (Weaviate, Pinecone, Milvus, pgvector), large-scale data processing and annotation pipelines.

MLOps / LLMOps

MLflow, Weights & Biases, Kubernetes, Docker, Ray Serve, Seldon, BentoML, model versioning, pipeline automation, CI/CD for ML, monitoring & observability, drift detection, data quality systems, guardrails, evaluation pipelines, GPU orchestration, autoscaling.

Software Engineering

Python, Java, Scala, C++, SQL, FastAPI, Flask, Node.js, REST/gRPC APIs, microservices, distributed systems, event-driven architectures, performance optimization, caching, serverless systems, cloud-native development.

Cloud & DevOps

AWS (SageMaker, Lambda, EKS, EMR, S3, DynamoDB), GCP (Vertex AI, GKE, BigQuery), Azure ML, Terraform, GitHub Actions, ArgoCD, CI/CD, logging/monitoring (Prometheus, Grafana), cost optimization.

PROFESSIONAL EXPERIENCE

Cohere — Generative AI Engineer

New York, NY | 2023–2025

Cohere builds enterprise-scale LLMs and AI platforms that power semantic search, RAG, and agentic workflows for global organizations.

At Cohere, I delivered production LLM systems, AI Agents, fine-tuning pipelines, and enterprise AI integrations that elevated model accuracy, safety, and automation impact.

- Led development of enterprise-grade LLM workflows (Cohere Command, Embed, Aya series) powering summarization, Q&A, and agentic automation across Fortune 100 clients, improving task success accuracy by ~27% and reducing prompt cost by 33% via optimized inference routing.
- Built multimodal text–audio–video embedding models using contrastive learning and dual-encoder architectures, enabling unified semantic search across documents, call transcripts, and video assets, resulting in 52% improvement in retrieval precision for enterprise knowledge bases.
- Designed a RAG 2.0 pipeline with structured retrieval, hybrid sparse+dense indexing, and graph-based context planning (Cohere Embed + Weaviate + pgvector), reducing hallucinations by 40% and doubling correctness for long-context enterprise queries.
- Created high-fidelity video-to-text generation and indexing pipelines using vision transformers, CLIP variants, SAM2, and temporal attention models; cut video analysis time from hours to minutes and boosted searchable video content coverage by 5×.
- Developed real-time speech-to-LLM pipelines using Whisper v3, MMS, and custom streaming attention mechanisms, achieving sub-150 ms end-to-end latency for multilingual voice assistants and increasing automation of voice workflows by 21%.
- Built a multimodal agent framework integrating text, audio, and video tool use with external API actions; improved agent task-completion rates from 63% → 88% for complex workflows such as compliance checks, support triage, and document analysis.
- Owned fine-tuning and PEFT pipelines (LoRA, QLoRA, DoRA, DPO, ORPO) for domain-specific LLM adaptation across finance, healthcare, and customer-service verticals, yielding 25–45% downstream accuracy gains on client benchmarks.
- Implemented production-ready LLM evaluation and safety systems using Ragas, DeepEval, automated red-teaming, and toxicity/PII classifiers, reducing unsafe output rates by >60% and establishing evaluation protocols adopted across 20+ client deployments.

- Drove multi-node distributed training (PyTorch FSDP, DeepSpeed, NeMo, Ray) for 7B–70B parameter models, optimizing throughput by 35% and reducing training cost per token by 22% through kernel fusion, mixed precision, and sequence packing strategies.
- Architected scalable LLM inference services using vLLM, TensorRT-LLM, Kubernetes, and Triton Inference Server, achieving p99 < 85 ms latency at scale and cutting GPU utilization cost by ~30% for enterprise deployments.
- Developed synthetic data generation pipelines using agentic workflows + evaluator models to produce high-quality text, audio, and video summaries, expanding training corpora by 10× while maintaining >93% evaluator-rated quality.
- Partnered with product, research, and enterprise clients to launch end-to-end generative AI solutions—including secure chat, multimodal knowledge assistants, and analytics copilots—driving \$12M+ in ARR attributed to new AI features built on Cohere’s platform.

LivePerson — Senior Machine Learning Engineer

New York, NY | 2020–2023

LivePerson is a conversational AI company providing intelligent messaging, automation, and real-time customer engagement for large enterprises.

I developed NLP and GenAI models, real-time ML systems, and customer-facing AI features that scaled to hundreds of millions of interactions monthly.

- Designed and deployed next-gen conversational NLP models (Transformers, BERT variants, DistilBERT, hierarchical intent models) across the Conversational Cloud, increasing intent-classification accuracy by 18% and lifting automated resolution rates by ~24% for enterprise clients.
- Engineered a large-scale semantic search + retrieval system using FAISS, Elasticsearch, and a PyTorch-based dual-encoder architecture to surface contextual replies with <80 ms latency, reducing average handling time (AHT) by 22% and saving >\$4M annually in operational costs.
- Built a multimodal conversation intelligence pipeline combining text embeddings, audio embeddings, and video-derived behavioral cues (OpenSMILE, Whisper, MediaPipe, OpenCV), enabling real-time insight extraction across chat, voice, and video channels, and boosting agent QA throughput by 33%.
- Led development of real-time voice-AI capabilities (ASR → NLU → response ranking) using Whisper, Kaldi, AWS Transcribe, and PyTorch, enabling LivePerson to expand automation to the voice channel with <200 ms end-to-end latency and reducing human handoff rates by 19%.
- Created a video-chat augmentation model leveraging a 3D CNN + attention mechanism to detect frustration, disengagement, and compliance signals, improving escalation prediction precision from 0.68 → 0.85 and increasing CSAT for premium customers by 9%.
- Built a personalized agent-assist recommender system using session embeddings, RNN-based context encoders, and retrieval-augmented re-ranking, increasing agent productivity by 23%, reducing response variance by ~17%, and speeding onboarding for new agents by 40%.
- Architected a high-throughput message-classification microservice (FastAPI, gRPC, Redis, Kubernetes, Triton Inference Server) processing >15k messages/sec at p99 <60 ms, powering automated routing, summarization, and sentiment detection for millions of conversations.
- Created a stream processing analytics pipeline (Kafka → Flink → Spark → S3) that analyzed billions of conversational events for anomalies, bot failures, and sentiment drift, cutting silent escalation incidents by 70% and improving platform reliability across Fortune 100 clients.
- Developed a weak-supervision + programmatic labeling pipeline (Snorkel, Prodigy, rule-based labeling functions) to expand NLP training data 10× while lowering annotation spend by 40%, accelerating model iteration cycles from weeks to days.

- Established robust ML observability and LLM evaluation workflows (MLflow, Prometheus, Ragas, custom drift detectors), reducing model degradation events by 55% and building a high-confidence deployment process for >25 production ML models.

Capital One — Data Engineer, AI/ML Platform

New York, NY | 2018–2020

Capital One is a technology-driven financial institution leveraging advanced ML and data platforms to power credit, fraud, and personalization systems.

I built ML infrastructure, streaming pipelines, and large-scale features enabling faster model deployment, improved risk accuracy, and real-time fraud detection.

- Engineered large-scale data pipelines (Spark, Scala, Python, EMR, Airflow) to ingest and transform billions of credit-card and transaction events daily, improving pipeline reliability from 92% → 99.7% across critical fraud and risk systems.
- Built production-grade feature engineering services powering ML models for credit risk, fraud detection, and customer decisioning, cutting model refresh time from 24 hours to under 3 hours and enabling near-real-time scoring.
- Designed and delivered a centralized Feature Store using Spark, Hive, and S3 that standardized 400+ reusable features across teams, reducing duplicate engineering work by 35% and boosting cross-team model consistency.
- Developed scalable model deployment and monitoring frameworks (Python, Docker, Kubernetes, MLflow) used by 20+ data science teams, accelerating model-to-production timelines by ~40%.
- Partnered with fraud and credit policy teams to productionize gradient boosting and logistic regression models, improving early fraud detection by ~12% and reducing false positives by ~9%.
- Created a real-time stream processing pipeline using Kafka + Spark Streaming to flag anomalous user behavior within sub-second latency, contributing to a \$18M annual reduction in fraud exposure.
- Implemented robust data quality validation workflows (schema checks, statistical drift, automated reconciliation) that decreased critical data incidents by 60%, significantly improving model stability and regulatory compliance.
- Optimized ETL jobs through partitioning, caching, and vectorized UDFs, cutting nightly batch processing costs by 25% and reducing compute time by ~50% for multiple enterprise-level jobs.
- Collaborated with security and governance teams to operationalize PII tokenization, encryption, and lineage tracking, ensuring ML pipelines remained fully compliant with PCI, GDPR, and internal risk controls.

Flatiron Health — Software Engineer

New York, NY | 2015–2018

Flatiron Health is a healthcare technology company transforming oncology care and research through large-scale EHR data and real-world evidence platforms.

I engineered backend systems, clinical data pipelines, and classical ML models that improved data quality, accelerated oncology analytics, and supported precision-care insights.

- Designed and implemented patient-level data ingestion pipelines for oncology EHR systems using Python, SQL, Spark, and Kafka, increasing structured-data coverage by ~42% and cutting manual abstraction effort by 30+ hours/week.

- Built production-grade microservices in Python/Flask and Java to process and validate clinical events (labs, medications, tumor progression), reducing downstream data discrepancies by 25% and improving report generation latency by ~40%.
- Developed classification and sequence-prediction ML models (logistic regression, gradient boosting, CRFs) to identify missing or inconsistent clinical documentation, raising abstractor recall from 0.82 → 0.93 and accelerating chart review cycles by ~18%.
- Engineered a disease-specific cohort builder leveraging custom ML-powered phenotyping algorithms, enabling researchers to generate real-world evidence cohorts 4× faster and improving cohort precision for rare cancers by ~15%.
- Partnered with oncology SMEs to build survival modeling pipelines using Kaplan–Meier, Cox PH, and time-to-event models, helping the RWE team produce regulatory-grade insights cited in multiple FDA submissions.
- Optimized ETL pipelines for OncoEMR data using distributed Spark jobs and columnar storage formats, improving job runtime by 55% and reducing monthly infrastructure costs by ~20%.
- Implemented automated data quality monitoring (drift detection, schema enforcement, clinical heuristics) and cut high-severity data issues by ~60%, strengthening trust in RWE datasets used by NCI-designated cancer centers.
- Collaborated cross-functionally with product, clinicians, and data scientists to launch ML-augmented features in Flatiron’s research platform, enabling pharma partners to interrogate >2M de-identified patient records with significantly improved filtering accuracy and performance.

EDUCATION

Massachusetts Institute of Technology

2013 – 2015 | Cambridge, MA

Master of Science – MS, Computer Science

Massachusetts Institute of Technology

2009 – 2013 | Cambridge, MA

Bachelor of Science – BS, Electrical Engineering and Computer Science (EECS)